Part Four

BEYOND SPACE AND TIME

I have illustrated the basis of quantum gravity, and the image of the world which emerges from it. In the final chapters I describe some consequences of the theory: what the theory tells us about phenomena such as the Big Bang and black holes. I also discuss the current state of possible experiments to test the theory, and what it seems to me nature is telling us – in particular with the failure of the expected observation of supersymmetric particles.

I conclude with a few reflections on what is still missing from our understanding of the world: especially thermodynamics, the role of information in a theory without time and space such as quantum gravity, and the re-emergence of time.

All of this takes us to the edge of what we know, to the vantage point from which we look upon what we definitely don't know, the immense mystery that surrounds us.